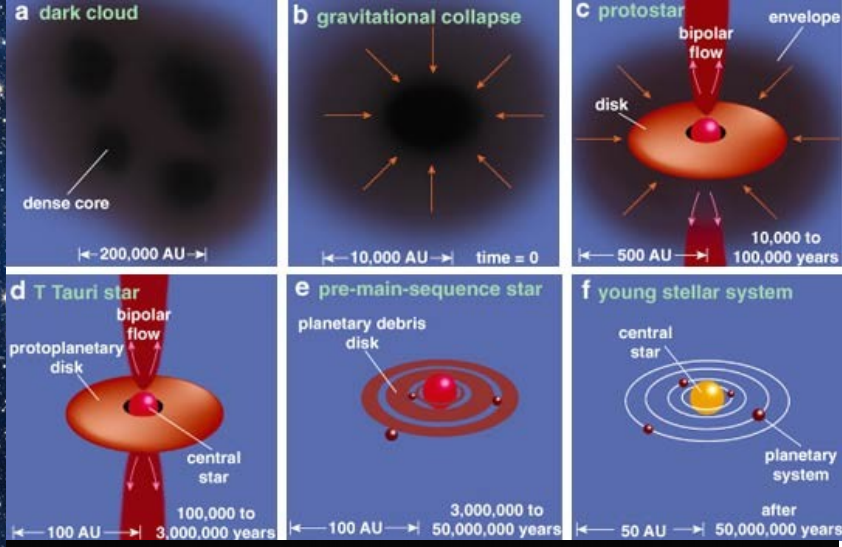
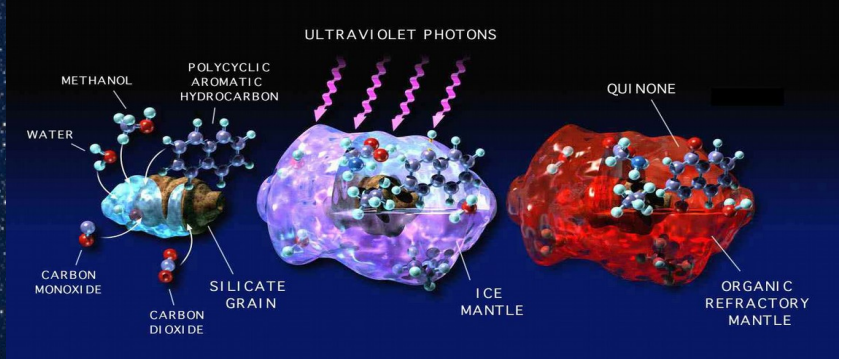
Stars getting
born


a dark cloud
dense core
200,000 AU
b gravitational collapse
10,000 AU
time = 0
c protostar
bipolar
flow
envelope
disk
500 AU
10,000 to
100,000 years
d T Tauri star
bipolar
flow
protoplanetary
disk
central
star
100 AU
100,000 to
3,000,000 years
e pre-main-sequence star
planetary debris
disk
100 AU
3,000,000 to
50,000,000 years
f young stellar system
central
star
planetary
system
50 AU
after
50,000,000 years


ULTRAVIOLET PHOTONS
METHANOL
POLYCYCLIC
AROMATIC
HYDROCARBON
WATER
QUINONE
CARBON
MONOXIDE
CARBON
DIOXIDE
SILICATE
GRAIN
ICE
MANTLE
ORGANIC
REFRACTORY
MANTLE